

Sleep Efficiency Classification Using Machine Learning

Project Final Report

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Abstract

Sleep quality is directly tied to lifestyle factors, yet the nonlinear relationships and patterns between these variables and sleep efficiency remains vague. In this project, we apply machine learning to classify sleep efficiency using the NHANES actigraphy dataset, which captures wrist-worn accelerometer measurements along with sleep timing and circadian rhythm metrics. We define good sleep quality as sleep efficiency of 0.85 or greater for the ratio of sleep time to total time in bed, and therefore convert this into a binary classification problem. Four models are trained and evaluated: linear perceptron, logistic regression, random forest, and a neural network. Features include circadian rhythm indicators, sleep-period acceleration, device-wear percentage, and timing. Our results show that the neural network achieves the highest performance with 62.2% accuracy and a F1 score of 0.695, while the random forest model follows closely behind (accuracy: 59.5%, F1: 0.659). Simple linear models perform near guessing level, suggesting that the data relationships are nonlinear. Feature importance analysis identifies circadian activity patterns and wrist acceleration as the strongest weighted model predictors of sleep efficiency. The class imbalance of the dataset limits recall on poor sleep cases across models, yet patterns between sleep and lifestyle features were derivable.

1 Introduction

1.1 Data Set

We are using the NHANES Sleep Data dataset. This CSV dataset contains approximately 85,000 rows and 42 columns, with a large set of features that influence one's sleep length and quality. Each row represents a single participant observation, and the columns include physiological and behavioral characteristics, such as sleep timing, wake timing, and activity intensity throughout the day and during the sleep period. This is an observational/real-world health dataset that shows how different habits and health variables affect sleep.

1.2 Machine Learning Models

1. Linear Perceptron

The linear perceptron is a simple linear classifier that attempts to find a separating hyperplane between the two classes. It works well when the data is linearly separable. This model will serve as a baseline for comparison.

2. Logistic Regression

Logistic regression is also a linear classifier, but produces probability outputs for each class. It is generally more stable than linear perceptron and performs better when classes overlap. It also allows us to interpret how individual features contribute to predictions.

3. Random Forest

Random forest is an ensemble method that constructs decision trees during training, with each decision tree being trained on a random subset of features and training samples. The final prediction is determined by a majority across all of the trees. This technique reduced overfitting compared to singular decision trees and can handle non-linear decision boundaries.

4. Neural Network

A neural network learns a composition of non-linear transformations through layers of weighted connections. During training, backpropagation adjusts the weights to minimize the prediction error. This allows the model to completely inspect feature interactions automatically without manual feature engineering.

1.3 Expected Outcomes

We expected lifestyle and behavioral factors from the NHANES dataset to show relationships with sleep efficiency. We predicted that physical activity patterns, circadian rhythms, and sleep timing would differ between those with good and poor sleep quality. We also expected more advanced models like Random Forest and Neural Networks to perform better than the simpler linear models because sleep behavior is nonlinear. Overall, our goal was to successfully classify sleep efficiency and understand which daily behaviors most affect it.

2 Related Work

1. Freeman et al., “Monitoring sleep duration, timing, and continuity among US youth and adults in NHANES using actigraphy”

Freeman et al. used the NHANES actigraphy data to characterize sleep duration, timing, and continuity among both youth and adults. This paper is useful to our project because it shows that the NHANES wearable data can be used to study sleep behavior rather than relying on self-reported sleep. It also provided us with the processed dataset we use for our project. Their work primarily focuses on population-level sleep patterns, most noticeably in how sleep duration and timing vary by age group. Without this paper and this study, we would not have the rich dataset we input into our training models for accurate predictions. This study truly breaks the barrier of studying sleep through its collection methods, and provided the foundation for our project to connect learning models to the sleep process and identify useful patterns.

2. Su et al., 2022 — “Epidemiology of accelerometer-based sleep parameters in US school-aged children and adults: NHANES 2011–2014”

Su et al. analyzes the NHANES accelerometer dataset we use and was developed in the Freeman paper. The study attempts to estimate sleep parameters such as sleep duration, onset time, and efficiency. They found that sleep duration had a parabolic correlation with age in that it hit its lowest point at around age 40. The study also derived that sleep efficiency generally decreased with age. This study was very similar in how it used its dataset to derive sleep patterns, but was much different in analytical methods. The paper hones in on epidemiological and statistical analysis, steering clear of machine learning methodology our project applies on the same dataset. We use the observations the authors procured, and take them to their limits with machine learning models.

3. Cai et al., 2025 — “Association between accelerometer-measured physical activity volume and sleep duration in older adults”

Cai et al. studied the relationship between accelerometer-measured physical activity volume and sleep duration in older adults. This paper is extremely relevant because it uses models including random forest, GBDT, XGBoost, LightGBM, and SHAP analysis to study sleep duration. GBDT (Gradient boosted decision trees) was their highest performing model, with an internal validation AUC (area under curve) of 0.859 and external validation AUC of 0.707. This is the most machine-learning related paper to our project. It demonstrates how machine learning can find nonlinear relationships between physical activity

and sleep. Our project is much broader in multiple domains. Physical activity is a subset of our data, and all age-ranges of patients are considered in this project's dataset. Differing models are also executed, as many of their models are higher performing yet were not taught in this course's curriculum.

4. Synthesis of Related Work

Combining the findings of these three related studies together, a clear progression from data collection to machine learning prediction on the patterns of sleep efficiency regarding lifestyle parameters. Freeman et al. processes the NHANES dataset and provides foundational knowledge, Su et al. shows that meaningful sleep patterns can be statistically analyzed from the dataset, and Cai et al indicates that machine learning can uncover complex relationships between activity and sleep measures. Our project further establishes the tailend of this progression by utilizing supervised learning methods to the NHANES sleep dataset in order to maximize prediction accuracy through pattern discovery and application. We expand past a single age-group or feature category and instead use broader categorical features to compare models we have learned in this course.

3 Methods

3.1 Data Pre-processing

1. Remove all unnecessary/redundant columns. These columns mostly contained identification variables and names that had no predictive value. There also contained multiple columns that contained the same information represented in different ways. Including them in the dataset would create unwanted noise that would affect the performance of our models. Removing these features allows the model to focus on more predictive features.
2. Change variables to type float. Our machine learning algorithms require numerical values to function properly. Some of the data in the dataset is represented as timestamps or strings. In order to use these variables, we must first convert the variable into a float type so we can use them in the algorithm.
3. Classifying **sleep_efficiency** ≥ 0.85 as "good sleep", otherwise, classify as "bad sleep". Originally, sleep_efficiency was a continuous variable from 0 to 1. By converting this variable into a binary classification, it allows us to frame this as a classification problem. This also gives us a useful evaluation metric which we can use to calculate accuracy, precision, recall, and F-1 scores.
4. Split the dataset 70/15/15 for training, testing, and validation respectively. The reason we chose this ratio is to prevent overfitting and to have more resistance against outliers while also giving a large enough sample for testing and validation.

3.2 Features

After preprocessing the data, and removing unnecessary features from the dataset, we ended with these features that we found were most important to determining the sleep quality of any given data point. Below is the list of features and what they represent. The descriptions of these features are from the NHANES Sleep Data Documentation

- **L5/L10:** Acceleration value for least active 5/10 hour period.
- **L5TIME/L10TIME:** Time start for the least active 5/10 hour period.
- **M5/M10:** Acceleration value for most active 5/10 hour period.

- **M5TIME/M10TIME:** Time start for the most active 5/10 hour period.
- **ACC_spt_sleep_mg:** Average acceleration asleep during sleep window.
- **sleeponset:** Time of sleep start.
- **wakeup:** Time of wake up.
- **dur_spt_min:** Duration of sleep window in minutes.
- **nonwear_perc_day:** Percentage of day where the bracelet was not worn.

3.3 Machine Learning Models

We evaluated four supervised classification models: linear perceptron, logistic regression, random forest, and neural network. All models were trained on the same preprocessed dataset for consistency.

Linear Perceptron: The perceptron is a linear classifier that updates its weights iteratively when misclassifications occur. It attempts to find a separating hyperplane between the two classes. Due to its simplicity and lack of probabilistic output, it serves as a baseline model.

Logistic Regression: Logistic regression models the probability of class membership using a sigmoid function. In our implementation, we used L2 regularization to prevent overfitting and set the maximum number of training iterations to 1000 to ensure convergence.

Random Forest: Random Forest is an ensemble learning method that creates multiple decision trees during training by taking a random subset of features. They then combine to produce a final prediction. We predict it will perform better than linear perceptron and logistic regression since it is good at generalizing data.

Neural Network: Neural networks are flexible, nonlinear models that contain multiple layers. Our implementation includes an input layer that contains all the features the model uses, two hidden layers, and an output of a binary classification. We predict that this model will perform the best because of its ability to handle non-linear datasets.

3.4 Hyperparameters

Each model required different types of hyper-parameters. They were chosen based on a variety of factors that are explained in the following.

1. Linear Perceptron:

The learning rate of the model was set to 0.1. This was large enough for the model to learn at a reasonable rate while being small enough to prevent oscillations, which caused a lot of instability during training. We ran the model for 100 epochs, which was enough to see convergence. We observed that adding more epochs didn't lead to any meaningful improvement in the model.

- Learning Rate: 0.1
- Epochs: 100

2. Logistic Regression:

We used L2 regularization to penalize large weights across all of the features, which prevented large weight from taking over the model. We used a standard C-value of 1.0 to shrink the weights while also not ignoring the regularization.

- Max Iterations: 1000
- Regularization: L2
- C: 1.0

3. Random Forest:

We found that setting the number of trees to 200 gave us enough trees to stabilize the variance of the model without making the computing cost too expensive. The max depth of the tree is set to 10 to prevent overfitting. Having minimum samples per leaf at 5 smooths the leaf predictions while also acting as a soft regularizer.

- # Trees: 200
- Max Depth: 10
- Minimum Samples Leaf: 5

4. Neural Network:

Our model has an architecture of $12 \rightarrow 64 \rightarrow 32 \rightarrow 1$. This allows us to input all 12 features required of the model and gives a single output for binary classification. The two hidden layers gives the model enough capacity to model non-linear relationships without over complicating the architecture given the size of our dataset. The activation function is ReLU which helps prevent the vanishing gradient problem. Values of dropout, batch normalization, and batch size were chosen as they were the most stable for the model. They helped stabilize the dataset, provide regularization, and prevent overfitting. Lastly, we set the epochs to 100 as the accuracy of the model started to plateau.

- Architecture: $12 \rightarrow 64 \rightarrow 32 \rightarrow 1$
- Activation Function: ReLU
- Dropout: 0.3
- Batch Normalization: After linear layers
- Epochs: 100
- Batch Size: 256

3.5 Evaluation Metrics

Model performance was evaluated using four standard classification metrics:

- **Accuracy:** Measures overall correctness of predictions.
- **Precision:** Measures how many predicted positive samples are actually correct.
- **Recall:** Measures how many actual positive samples are correctly identified.
- **F1 Score:** The mean of precision and recall, providing a balance between the two.

Accuracy alone can be misleading, so precision, recall, and F1 score provide a more complete understanding of model performance.

Results	Linear Perceptron	Logistic Regression	Random Forest	Neural Network
Accuracy	0.516	0.5658	0.5950	0.6220
Precision	0.6669	0.7225	0.7424	0.7446
Recall	0.4991	0.5568	0.5931	0.6515
F-1	0.5709	0.6289	0.6594	0.6949

Table 1: Model performance comparison

4 Experimental Results

Table 1 presents the binary classification performance of the four models: Linear Perceptron, Logistic Regression, Random Forest, and Neural Network. The evaluation of this model is instrumented through accuracy, precision, recall, and F-1 score. The neural network achieved the highest scores in all four metrics (accuracy: 0.6220, precision: 0.7446, recall: 0.6515, F-1 score: 0.6949) making it the best performing model. Random Forest was a competitive second, having particularly similar precision and accuracy scores. The Linear Perceptron model performed the worst across all metrics with an accuracy of 0.516.

4.1 Model Breakdown

1. Linear Perceptron Analysis

The linear perceptron model iteratively adjusts it’s weight for each misclassified sample. It updates weight with a learning rate of 0.1 and goes through 100 epochs. As this was our baseline model, it was also our simplest. The model achieved an accuracy of 0.5160, a precision of 0.6669, a recall of 0.4991, and a F-1 of 0.5709.

The model performed the worst out of all 4 models in every aspect. With an accuracy score of 0.516, the model didn’t perform much better than randomly guessing. This suggests that the features of sleep quality in this dataset are not linearly separable. This is also supported by the low recall score, which correctly predicted good sleep less than half of the time.

As seen in Table 3, among the most predictive features were M5TIME, M5VALUE, and M10VALUE, as well as the percentage of the day the device wasn’t worn, and the wake up time. According to the model, this suggests that the most active time during the sleep period was the most important feature. This also lines up with our original prediction that circadian activity would be the biggest influence on sleep quality.

Feature	Coefficient
% sleep device not worn	+0.54
sleeponset	+0.28
M10VALUE	+0.21
wakeup	-0.30
% day device not worn	-0.46

Table 2: Top Logistic Regression Features

Feature	Coefficient
% of day device worn	+0.5446
Wake up time	-0.5076
M5TIME	-0.4898
M5VALUE	0.4716
M10VALUE	-0.3584

Table 3: Top Linear Perceptron Features

2. Logistic Regression Analysis

The logistic regression model was trained using L2 regularization with a maximum of 1000 iterations and balanced class weighting to account for the dataset’s 2:1 good to poor sleep efficiency ratio. The model achieved an accuracy of 0.5658, precision of 0.7225, recall of 0.5568, and an F-1 score of 0.6289.

Compared to the baseline linear perceptron, logistic regression showed a small improvement across all metrics. This shows that the sigmoid-based classification and regularization helped better separate the classes than the linear perceptron model. However, random forest and neural network still outperformed it in every category. This suggests that while logistic regression captured some useful linear relationships, it was limited when it came to the nonlinear interactions in the sleep behavior data. The model demonstrated relatively strong precision, meaning that when it predicted good sleep, it was often right. However, recall remained lower, indicating that it still struggled to identify good sleep samples and was sensitive to class imbalance.

As seen in Table 2, the most predictive features were percent of sleep non-wear, sleep onset time, M10VALUE, wake-up time, and percent of day non-wear. Positive coefficients for sleep onset and M10VALUE suggest that stronger circadian rhythm and sleep timing patterns were associated with improved sleep classification, while negative coefficients for wake up time suggest poorer sleep outcomes. These results line up with our original prediction that circadian activity and sleep timing would influence sleep efficiency the most.

3. Random Forest Analysis

The random forest model was constructed with 200 estimators, a maximum tree depth of 10, and a minimum of 5 samples per leaf. Since the dataset had a 2:1 good to poor sleep efficiency ratio, we applied a balanced class weight to prevent the model from overpredicting good sleep. The model achieved an accuracy of 0.5950, precision of 0.7424, recall of 0.5931, and an F-1 score of 0.6594.

The model demonstrated a false negative rate of 41%, meaning the model failed to identify around 4000 cases of good sleep and instead labeled them with poor sleep efficiency. Whereas when the model predicted good sleep efficiency, it was correct nearly 74% of the time. This suggests that the model is conservative and reluctant to predict good sleep quality unless the actigraphy signal was strong.

As seen in Table 4, feature importance indicates that sleep period duration was the best predictive variable, with a model weight of 19.7%. The least active 5-hour window intensity (14.8%) and in-sleep accelerometry (13.1%) were also prominent, along with circadian ratio (M5VALUE/L5VALUE) and device wear time. In contrast, day of week (1.8%) and percentage of day without the device (1.4%) contributed little to prediction, suggesting that weekday patterns and study compliance were not associated with good sleep.

Feature	Coefficient
Sleep period duration	0.197
L5VALUE	0.148
Wrist Acceleration	0.131
Circadian Ratio (peak/rest act)	0.108
Device Wear Time	0.093

Table 4: Random Forest Top Features

Feature	Coefficient
L5VALUE	0.7713
Wrist Acceleration during sleep	0.6821
Time in bed	0.4081
L10VALUE	0.1412
M10VALUE	0.1025

Table 5: Neural Network Top Features

4. Neural Network Analysis

The neural network was implemented using a multilayer perceptron (MLP) with two hidden layers (12, 64, 32, 1). Each of the hidden layers used batch normalization and dropout rate of 0.3 to stabilize the training and reduce overfitting. The model was trained for 100 epochs using the Adam optimizer, and we halved learning rate when validation loss flatlines. We combined binary cross entropy and sigmoid in BCEWithLogitsLoss to even out the class frequency. We then restored the best model checkpoint before evaluation. This resulted in the neural network achieving the best performance in all metrics, with an accuracy of 0.6220, precision 0.7446, recall 0.6515, and F-1 score of 0.6949. The training and validation loss curves converged smoothly, showing that dropout was effective in preventing overfitting. The confusion matrix portrayed a false negative rate of 34.8%, which was much lower than the random forest model. The neural network improved marginally in recall score, the primary reason that it leads in the F-1 score metric.

Using SHAP analysis, the weights of the neural network were inspectable and can be found in Table 5. It revealed that the L5VALUE (least active 5 hour window intensity) and in-sleep accelerometry were crucial in binary prediction, accounting for a majority of the SHAP value with 0.7713 and 0.6821 weights. Time in bed has a weight of 0.4081, while L10VALUE and M10VALUE (least and most active 10 hour intensities) had weights of 0.1412 and 0.1025. L5TIME, M5TIME, and circadian ratio were among the least prominent factors. This suggests that for predicting sleep efficiency, the intensity of rest and activity matters starkly more than when those periods happened.

4.2 Discussion of Results

Overall, our results showed that sleep efficiency can be predicted using actigraphy-based lifestyle and behavioral data. Simpler models like linear perceptron and logistic regression were able to find some useful patterns, but random forest and neural network models performed better. This suggests that sleep quality is influenced by more complex, nonlinear relationships that advanced models can better handle.

Across all models, circadian rhythm and activity-based features were some of the strongest predictors of sleep quality. This supports our original prediction that these behaviors would most impact sleep efficiency.

One major challenge was class imbalance, since the dataset contained more good sleep samples than poor sleep samples. This made it harder for models to accurately predict poor sleep cases. In addition, our project mainly used actigraphy-based features, so other important influences weren't included and may have been missed.

5 Conclusion

This project showed that sleep efficiency can be predicted using actigraphy-based lifestyle and behavioral data from the NHANES Sleep dataset. Daily habits like physical activity, sleep timing, and circadian rhythm patterns were important in predicting good or poor sleep quality. These features show that daily movement and rest patterns are closely connected to sleep health.

Our results also showed that sleep quality is influenced by more complex relationships that simple linear models can't fully understand. While linear perceptron and logistic regression were good baselines, Random Forest and Neural Network models performed better overall. This suggests that more advanced models are better at identifying the nonlinear patterns between lifestyle behaviors and sleep efficiency.

However, there were still some limitations. The dataset had more good sleep samples than poor sleep samples, which made it harder for models to accurately predict poor sleep cases. In addition, our project only used actigraphy-based features, so it may be missing other important factors like diet, stress, or medical conditions that also affect sleep.

Overall, this project demonstrates that machine learning can be used for analyzing sleep health and identifying important lifestyle factors related to sleep quality. Future improvements could include using more diverse health data, better handling class imbalance, and testing more advanced models to improve prediction accuracy.

6 References

- Cai, XiaoTao, et al. “Association between Accelerometer-Measured Physical Activity Volume and Sleep Duration in Older Adults: A Cross-Sectional Interpretable Machine Learning Analysis.” *Frontiers in Public Health*, vol. 13, 2025, article 1635020, doi:10.3389/fpubh.2025.1635020.
- Freeman, Joshua R., et al. “Monitoring Sleep Duration, Timing, and Continuity among US Youth and Adults in NHANES Using Actigraphy.” *Sleep Health*, vol. 12, no. 1, 2026, pp. 104–112, doi:10.1016/j.sleh.2025.10.008.
- Su, Shaoyong, et al. “Epidemiology of Accelerometer-Based Sleep Parameters in US School-Aged Children and Adults: NHANES 2011–2014.” *Scientific Reports*, vol. 12, no. 1, 2022, article 7680, doi:10.1038/s41598-022-11848-8.